DT059A - Applied Optimization
Lab 2 Report

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1. Introduction

In the second lab of the Applied Optimization course, I explored constrained and unconstrained optimization problems using the Karush–Kuhn–Tucker (KKT) framework. The main objective of this lab was to formulate optimization problems in standard form, derive and solve the corresponding KKT conditions, and analyze the role of active and inactive constraints. Additionally, graphical interpretations were made using curve plots to support the analytical results and develop insights regarding feasibility, optimality and unconstrained conditions.

2. Experimental Setup

Throughout the lab, all implementations were developed using Python. Symbolic computations for deriving gradients and KKT conditions were performed with the `SymPy` library, while numerical evaluations and visualizations were carried out using `NumPy` and `Matplotlib`.

Problems 1–4 were implemented within a single script since they share a similar two-dimensional structure and involve comparable KKT formulations with either inequality or equality constraints. In contrast, the remaining problems were solved in separate scripts due to differences in dimensionality, constraint structure, or problem type (unconstrained or higher-dimensional cases).

3. Solutions

3.1. Problem - 1 KKT analysis and solution

$$\max_{x_1,x_2} \quad F(x_1,x_2) = 4x_1^2 + 3x_2^2 - 5x_1x_2 - 8 \tag{1}$$
$$\text{subject to} \quad x_1 + x_2 \le 4. \tag{2}$$

I convert to minimization (Standard form) and write the inequality as $g(x) \le 0$:

$$\min_{x_1,x_2} \quad f(x_1,x_2) = -F(x_1,x_2) = -4x_1^2 - 3x_2^2 + 5x_1x_2 + 8, \tag{3}$$
$$\text{subject to.} \quad g(x_1,x_2) = x_1 + x_2 - 4 \le 0. \tag{4}$$

Lagrangian:

$$\mathcal{L}(x_1,x_2,\lambda) = f(x_1,x_2) + \lambda\, g(x_1,x_2), \qquad \lambda \ge 0. \tag{5}$$

KKT conditions:

$$\text{Stationarity:} \quad \frac{\partial \mathcal{L}}{\partial x_1} = -8x_1 + 5x_2 + \lambda = 0, \tag{6}$$
$$\frac{\partial \mathcal{L}}{\partial x_2} = 5x_1 - 6x_2 + \lambda = 0, \tag{7}$$
$$\text{Primal feasibility:} \quad x_1 + x_2 - 4 \le 0, \tag{8}$$
$$\text{Dual feasibility:} \quad \lambda \ge 0, \tag{9}$$
$$\text{Complementary slackness:} \quad \lambda(x_1 + x_2 - 4) = 0. \tag{10}$$

Active-set candidates:

- *Inactive constraint* ($x_1 + x_2 - 4 < 0 \Rightarrow \lambda = 0$): From (6)–(7) with $\lambda = 0$, we obtain
$$(x_1,x_2) = (0,0),$$
which is feasible since $0 \le 4$. The slack is $s = 4 - (x_1 + x_2) = 4$ (thus $s^2 = 16$).
- *Active constraint* ($x_1 + x_2 - 4 = 0$): Solving (6)–(7) together with $x_1 + x_2 = 4$ yields
$$(x_1,x_2) = \left(\frac{11}{6}, \frac{13}{6}\right), \qquad \lambda = \frac{23}{6} > 0.$$
Here the slack is $s = 0$ (thus $s^2 = 0$).

Objective values at KKT candidates:

$$F(0,0) = -8, \qquad F\left(\frac{11}{6}, \frac{13}{6}\right) = -\frac{1}{3}.$$

Even though the points above satisfy the KKT conditions, the original maximization problem has *no finite maximizer*. Indeed, consider the feasible ray $(x_1,x_2) = (-t,0)$ with $t \ge 0$, which satisfies $x_1 + x_2 = -t \le 4$. Along this ray,

$$F(-t,0) = 4t^2 - 8 \xrightarrow[t\to\infty]{} +\infty,$$

therefore the objective is unbounded above. Hence, the problem is *unbounded* and does not admit an optimal (finite) maximum.

In Fig. 1 the feasible set is the half-plane below the line $x_1 + x_2 = 4$. The two marked KKT points correspond to (i) an interior stationary point $(0,0)$ with inactive constraint and (ii) a boundary stationary point $\left(\frac{11}{6}, \frac{13}{6}\right)$ with active constraint. However, the level curves show that moving in a feasible direction toward large negative x_1 (e.g., along the arrow) reaches higher and higher objective values, confirming unboundedness.

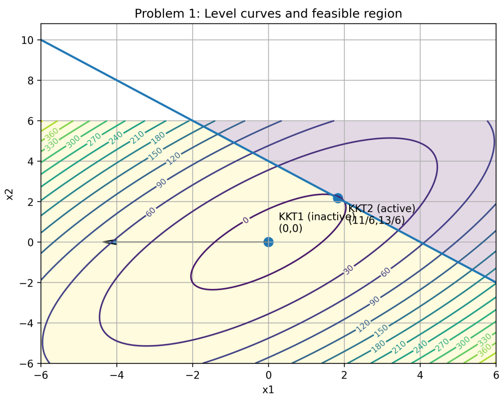

Figure 1. Level curves of $F(x_1,x_2)$, feasible region $x_1 + x_2 \le 4$, and KKT candidates for Problem 1.

3.2. Problem - 2 KKT analysis and solution

$$\max_{x_1,x_2} \quad F(x_1,x_2) = 4x_1^2 + 3x_2^2 - 5x_1x_2 - 8x_1 \tag{11}$$
$$\text{subject to.} \quad x_1 + x_2 \le 4. \tag{12}$$

I convert to minimization (Standard form) and write the inequality as $g(x) \le 0$:

$$\min_{x_1,x_2} \quad f(x_1,x_2) = -F(x_1,x_2) = -4x_1^2 - 3x_2^2 + 5x_1x_2 + 8x_1, \tag{13}$$
$$\text{subject to.} \quad g(x_1,x_2) = x_1 + x_2 - 4 \le 0. \tag{14}$$

Lagrangian:

$$\mathcal{L}(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda\, g(x_1, x_2), \qquad \lambda \geq 0. \tag{15}$$

KKT conditions:

$$\begin{aligned}
\text{Stationarity:} \quad & \frac{\partial \mathcal{L}}{\partial x_1} = -8x_1 + 5x_2 + 8 + \lambda = 0, & (16)\\
& \frac{\partial \mathcal{L}}{\partial x_2} = 5x_1 - 6x_2 + \lambda = 0, & (17)\\
\text{Primal feasibility:} \quad & x_1 + x_2 - 4 \leq 0, & (18)\\
\text{Dual feasibility:} \quad & \lambda \geq 0, & (19)\\
\text{Complementary slackness:} \quad & \lambda(x_1 + x_2 - 4) = 0. & (20)
\end{aligned}$$

Active-set candidates:

- *Inactive constraint* ($x_1 + x_2 - 4 < 0 \Rightarrow \lambda = 0$): From (16)–(17) with $\lambda = 0$, we obtain

  $$(x_1, x_2) = (-1, -1),$$

  which is feasible since $-2 \leq 4$. The slack is $s = 4 - (x_1 + x_2) = 6$ (thus $s^2 = 36$).
- *Active constraint* ($x_1 + x_2 - 4 = 0$): Solving (16)–(17) together with $x_1 + x_2 = 4$ yields

  $$(x_1, x_2) = (2, 2), \qquad \lambda = 6 > 0.$$

  Here the slack is $s = 0$ (thus $s^2 = 0$).

Objective values at KKT candidates:

$$F(-1, -1) = 8, \qquad F(2, 2) = 4.$$

Although the KKT conditions are satisfied at the candidate points, the maximization problem does not admit a finite optimal solution. This is because the objective function can be increased indefinitely while remaining feasible. For example, along the feasible direction $(x_1, x_2) = (-t, 0)$ with $t \geq 0$, the constraint $x_1 + x_2 \leq 4$ is always satisfied, while the objective value

$$F(-t, 0) = 4t^2 + 8t$$

grows without bound as t increases. Therefore, the problem is unbounded and has no finite maximizer.

In Fig. 2, the feasible region is again the half-plane below the line $x_1 + x_2 = 4$. The KKT points correspond to an interior stationary point $(-1, -1)$ with inactive constraint and a boundary stationary point $(2, 2)$ with active constraint. The level curves show that moving along feasible directions toward large negative x_1 leads to arbitrarily large objective values, confirming the unbounded nature of the maximization problem.

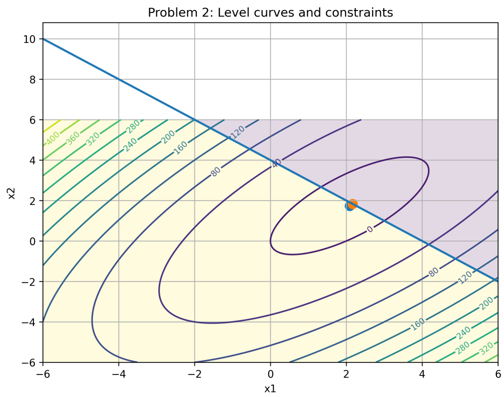

Figure 2. Level curves of $F(x_1, x_2)$, feasible region $x_1 + x_2 \leq 4$, and KKT candidates for Problem 2.

3.3. Problem - 3 KKT analysis and solution

$$\min_{x_1, x_2} \quad f(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 1)^2 \tag{21}$$

$$\begin{aligned}
\text{subject to.} \quad & x_1 + x_2 \geq 4, & (22)\\
& x_1 - x_2 - 2 = 0. & (23)
\end{aligned}$$

The inequality constraint is rewritten in standard form as $g(x) \leq 0$:

$$g(x_1, x_2) = 4 - x_1 - x_2 \leq 0, \tag{24}$$

and the equality constraint is written as

$$h(x_1, x_2) = x_1 - x_2 - 2 = 0. \tag{25}$$

Lagrangian:

$$\mathcal{L}(x_1, x_2, \lambda, \mu) = f(x_1, x_2) + \lambda\, g(x_1, x_2) + \mu\, h(x_1, x_2), \qquad \lambda \geq 0. \tag{26}$$

KKT conditions:

$$\begin{aligned}
\text{Stationarity:} \quad & \frac{\partial \mathcal{L}}{\partial x_1} = 2(x_1 - 1) - \lambda + \mu = 0, & (27)\\
& \frac{\partial \mathcal{L}}{\partial x_2} = 2(x_2 - 1) - \lambda - \mu = 0, & (28)\\
\text{Primal feasibility:} \quad & 4 - x_1 - x_2 \leq 0, & (29)\\
& x_1 - x_2 - 2 = 0, & (30)\\
\text{Dual feasibility:} \quad & \lambda \geq 0, & (31)\\
\text{Complementary slackness:} \quad & \lambda(4 - x_1 - x_2) = 0. & (32)
\end{aligned}$$

From the equality constraint (30), we obtain

$$x_2 = x_1 - 2.$$

Substituting into the inequality constraint gives

$$x_1 + x_2 = 2x_1 - 2 \geq 4 \quad \Rightarrow \quad x_1 \geq 3.$$

Inactive inequality ($4 - x_1 - x_2 < 0 \Rightarrow \lambda = 0$): Using $\lambda = 0$ in (27)–(28) yields

$$2(x_1 - 1) + \mu = 0, \qquad 2(x_2 - 1) - \mu = 0.$$

Solving together with $x_2 = x_1 - 2$ gives

$$(x_1, x_2) = (3, 1), \qquad \mu = -4.$$

This point satisfies $x_1 + x_2 = 4$, hence the inequality constraint is actually *active*. Therefore, this case is inconsistent.

Active inequality ($4 - x_1 - x_2 = 0$): Solving (27)–(28) together with

$$x_1 + x_2 = 4, \qquad x_1 - x_2 = 2,$$

yields the unique KKT point

$$(x_1, x_2) = (3, 1), \qquad \lambda = 0, \qquad \mu = -4.$$

Objective value at the KKT point:

$$f(3, 1) = (3 - 1)^2 + (1 - 1)^2 = 4.$$

Since the objective function is strictly convex and the feasible set defined by a line and a half-plane is convex, this KKT point is the *unique global minimizer* of the problem.

In Fig. 3, the feasible set is the intersection of the half-plane $x_1 + x_2 \geq 4$ and the line $x_1 - x_2 = 2$. The minimum is attained at their intersection point $(3, 1)$, where the smallest level curve of the objective function touches the feasible region.

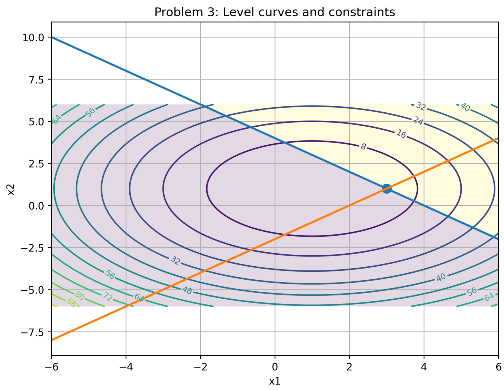

Figure 3. Level curves of $f(x_1, x_2)$, constraints $x_1 + x_2 \geq 4$ and $x_1 - x_2 = 2$, and the KKT solution for Problem 3.

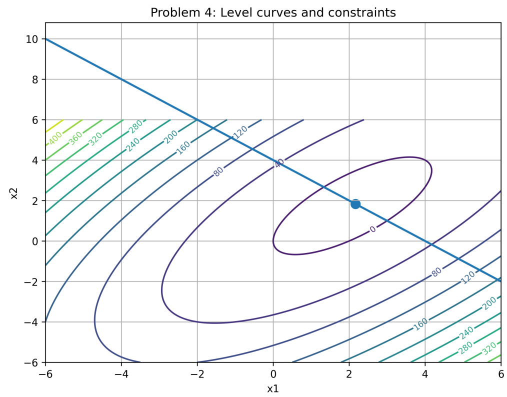

Figure 4. Level curves of $f(x_1, x_2)$, constraint $x_1 + x_2 = 4$, and the optimal solution for Problem 4.

3.4. Problem - 4 KKT analysis and solution

$$\min_{x_1,x_2} \quad f(x_1, x_2) = 4x_1^2 + 3x_2^2 - 5x_1x_2 - 8x_1 \tag{33}$$

$$\text{subject to.} \quad x_1 + x_2 = 4. \tag{34}$$

The equality constraint is written in standard form as

$$h(x_1, x_2) = x_1 + x_2 - 4 = 0. \tag{35}$$

Lagrangian:

$$\mathcal{L}(x_1, x_2, \mu) = f(x_1, x_2) + \mu\, h(x_1, x_2). \tag{36}$$

KKT conditions:

$$\text{Stationarity:} \quad \frac{\partial \mathcal{L}}{\partial x_1} = 8x_1 - 5x_2 - 8 + \mu = 0, \tag{37}$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 6x_2 - 5x_1 + \mu = 0, \tag{38}$$

$$\text{Primal feasibility:} \quad x_1 + x_2 - 4 = 0. \tag{39}$$

Solving (37)–(38) together with $x_1 + x_2 = 4$ yields

$$(x_1, x_2) = \left(\frac{25}{12}, \frac{23}{12}\right), \qquad \mu = -\frac{3}{4}.$$

Objective value at the optimum:

$$f\left(\frac{25}{12}, \frac{23}{12}\right) = 4\left(\frac{25}{12}\right)^2 + 3\left(\frac{23}{12}\right)^2 - 5\left(\frac{25}{12}\right)\left(\frac{23}{12}\right) - 8\left(\frac{25}{12}\right) = -\frac{145}{48}.$$

Since the objective function is convex quadratic (positive definite Hessian) and the constraint is affine, this stationary point is the *unique global minimizer*.

In Fig. 4, the feasible set is the line $x_1 + x_2 = 4$. The minimum is attained at the point $\left(\frac{25}{12}, \frac{23}{12}\right)$ where a level curve of f is tangent to the constraint line.

3.5. Problem - 5 KKT analysis and solution

$$\min_{x_1,x_2,x_3} \quad f(x_1, x_2, x_3) = (x_1 - 2)^2 + (x_2 - 1)^2 + (x_3 - 3)^2 \tag{40}$$

$$\text{subject to.} \quad x_1 + x_2 + x_3 \geq 6, \tag{41}$$

$$x_1 - 2x_2 \leq 1, \tag{42}$$

$$x_3 \geq 1. \tag{43}$$

I rewrite all inequalities in standard form $g_i(x) \leq 0$:

$$g_1(x) = 6 - x_1 - x_2 - x_3 \leq 0, \tag{44}$$

$$g_2(x) = x_1 - 2x_2 - 1 \leq 0, \tag{45}$$

$$g_3(x) = 1 - x_3 \leq 0. \tag{46}$$

Lagrangian:

$$\mathcal{L}(x_1, x_2, x_3, \lambda) = f(x_1, x_2, x_3) + \lambda_1 g_1(x) + \lambda_2 g_2(x) + \lambda_3 g_3(x), \qquad \lambda_i \geq 0. \tag{47}$$

KKT conditions:

$$\text{Stationarity:} \quad \begin{aligned} &\nabla f(x) + \lambda_1 \nabla g_1(x) \\ &\quad + \lambda_2 \nabla g_2(x) \\ &\quad + \lambda_3 \nabla g_3(x) = 0, \end{aligned} \tag{48}$$

$$\text{Primal feasibility:} \quad g_i(x) \leq 0, \quad i = 1, 2, 3, \tag{49}$$

$$\text{Dual feasibility:} \quad \lambda_i \geq 0, \quad i = 1, 2, 3, \tag{50}$$

$$\text{Complementary slackness:} \quad \lambda_i g_i(x) = 0, \quad i = 1, 2, 3. \tag{51}$$

Since the problem contains three inequality constraints, all possible combinations of active and inactive constraints were considered. Each constraint was assumed to be either active ($g_i(x) = 0$) or inactive ($\lambda_i = 0$), resulting in $2^3 = 8$ possible active-set configurations.

The unconstrained minimizer of f is clearly $(2, 1, 3)$, and it satisfies all constraints:

$$\begin{aligned} x_1 + x_2 + x_3 &= 2 + 1 + 3 = 6 && (\text{active } g_1 = 0), \\ x_1 - 2x_2 &= 2 - 2 = 0 \leq 1 && (\text{inactive}), \\ x_3 &= 3 \geq 1 && (\text{inactive}). \end{aligned}$$

Thus, the KKT solution is

$$x^\star = (2, 1, 3), \qquad \lambda_1 = \lambda_2 = \lambda_3 = 0,$$

with slack values $s_1 = 0$, $s_2 = 1 - (x_1 - 2x_2) = 1$, and $s_3 = x_3 - 1 = 2$ (equivalently $s_i = -g_i(x^\star)$ for $g_i \leq 0$). The objective value is

$$f(x^\star) = 0.$$

Since f is a strictly convex quadratic function and the feasible set (intersection of half-spaces) is convex, this KKT point is the unique global minimizer.

In Fig. 5, the plotted points represent sampled feasible solutions in 3D. The optimum $x^\star = (2, 1, 3)$ lies on the active plane $x_1 + x_2 + x_3 = 6$ while the other constraints remain slack, which is consistent with the KKT multipliers being zero.

3.6. Problem - 6 KKT analysis and solution

$$\min_{x \in \mathbb{R}} f(x) = x^4 - 2x^2 \qquad (\text{unconstrained}). \tag{52}$$

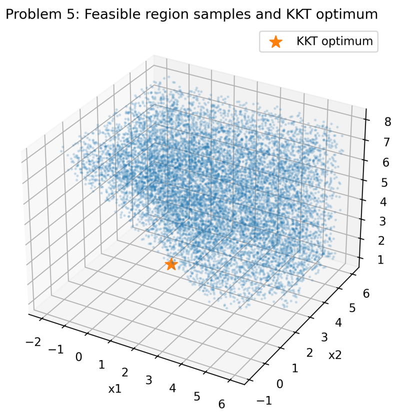

Figure 5. Sampled feasible region in (x_1, x_2, x_3) and the KKT optimum for Problem 5.

Since the problem is unconstrained, the KKT condition reduces to the *stationarity* condition $f'(x) = 0$. We compute:

$$f'(x) = 4x^3 - 4x = 4x(x^2 - 1), \tag{53}$$
$$f''(x) = 12x^2 - 4. \tag{54}$$

Stationary points: Solving $f'(x) = 0$ gives

$$x \in \{-1, 0, 1\}.$$

Classification (second-order test):

- At $x = 0$: $f''(0) = -4 < 0$, hence $x = 0$ is a *local maximum*. Also, $f(0) = 0$.
- At $x = \pm 1$: $f''(\pm 1) = 12 - 4 = 8 > 0$, hence $x = \pm 1$ are *local minima*. Moreover,

$$f(\pm 1) = 1 - 2 = -1.$$

Because $f(x) \to +\infty$ as $|x| \to \infty$, the local minima at $x = \pm 1$ are also the *global minimizers*, and the global minimum value is -1.

Key takeaway (stationarity is not sufficient): All three points $x \in \{-1, 0, 1\}$ satisfy the stationarity (KKT) condition $f'(x) = 0$, but only $x = \pm 1$ are minima while $x = 0$ is a local maximum. This illustrates that in a *nonconvex* problem, satisfying the KKT (stationarity) condition alone is *not sufficient* to guarantee optimality.

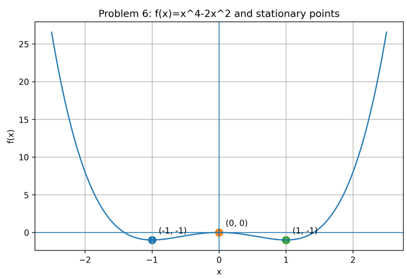

Figure 6. Plot of $f(x) = x^4 - 2x^2$ showing stationary points at $x = -1, 0, 1$. The point $x = 0$ is a local maximum, whereas $x = \pm 1$ are (global) minima.

3.7. Problem - 7 KKT analysis and solution

$$\min_{x_1, x_2} \quad f(x_1, x_2) = (x_1 - 3)^2 + (x_2 - 2)^2 \tag{55}$$
$$\text{subject to.} \quad x_1 \geq 2, \tag{56}$$
$$x_2 \geq 1, \tag{57}$$
$$x_1 + x_2 \leq 4. \tag{58}$$

I rewrite inequalities in standard form $g_i(x) \leq 0$:

$$g_1(x) = 2 - x_1 \leq 0, \tag{59}$$
$$g_2(x) = 1 - x_2 \leq 0, \tag{60}$$
$$g_3(x) = x_1 + x_2 - 4 \leq 0. \tag{61}$$

Lagrangian:

$$\mathcal{L}(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda_1 g_1(x) + \lambda_2 g_2(x) + \lambda_3 g_3(x), \qquad \lambda_i \geq 0. \tag{62}$$

KKT conditions:

$$\text{Stationarity:} \quad \frac{\partial \mathcal{L}}{\partial x_1} = 2(x_1 - 3) - \lambda_1 + \lambda_3 = 0, \tag{63}$$
$$\frac{\partial \mathcal{L}}{\partial x_2} = 2(x_2 - 2) - \lambda_2 + \lambda_3 = 0, \tag{64}$$
$$\text{Primal feasibility:} \quad g_i(x) \leq 0, \ i = 1, 2, 3, \tag{65}$$
$$\text{Dual feasibility:} \quad \lambda_i \geq 0, \ i = 1, 2, 3, \tag{66}$$
$$\text{Complementary slackness:} \quad \lambda_i g_i(x) = 0, \ i = 1, 2, 3. \tag{67}$$

The unconstrained minimizer of f is $(3, 2)$, but it violates $x_1 + x_2 \leq 4$ since $3 + 2 = 5$. Therefore, the solution must lie on the boundary $x_1 + x_2 = 4$ while also satisfying $x_1 \geq 2$ and $x_2 \geq 1$.

Solving the KKT system gives the optimal point

$$x^\star = \left(\frac{5}{2}, \frac{3}{2}\right), \qquad f(x^\star) = \left(\frac{5}{2} - 3\right)^2 + \left(\frac{3}{2} - 2\right)^2 = \frac{1}{2}.$$

Active/inactive constraints: At $x^\star$:

$$g_1(x^\star) = 2 - \frac{5}{2} = -\frac{1}{2} < 0 \quad \text{(inactive)}, \qquad g_2(x^\star) = 1 - \frac{3}{2} = -\frac{1}{2} < 0 \quad \text{(inactiv}$$

Thus, by complementary slackness, $\lambda_1 = \lambda_2 = 0$. Using stationarity (63)–(64) at $x^\star$ yields $\lambda_3 = 1$.

Slack and squared slack: $s_i = -g_i(x^\star) \geq 0$ (since $g_i \leq 0$):

$$s_1 = \frac{1}{2}, \ s_1^2 = \frac{1}{4}, \qquad s_2 = \frac{1}{2}, \ s_2^2 = \frac{1}{4}, \qquad s_3 = 0, \ s_3^2 = 0.$$

Since the objective is strictly convex quadratic and the feasible region is a convex polytope, this KKT point is the unique global minimizer.

In Fig. 7, the feasible set is the intersection of the half-planes $x_1 \geq 2$, $x_2 \geq 1$, and $x_1 + x_2 \leq 4$. The optimum $x^\star = (2.5, 1.5)$ lies on the active boundary line $x_1 + x_2 = 4$, while the other two constraints remain inactive (positive slack), consistent with $\lambda_3 > 0$ and $\lambda_1 = \lambda_2 = 0$.

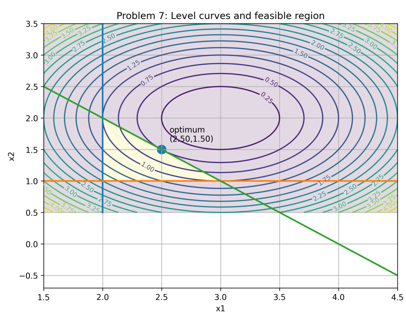

Figure 7. Level curves of $f(x_1, x_2)$, feasible region, and the KKT optimum for Problem 7.

4. Appendix: Code Snippets

All implementations are done locally in Python using PyCharm and are available on GitHub: https://github.com/kaanoztekin99/applied-optimization